PET ADOPTION AND RELATED SERVICES APPLICATION

A Project Report

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Under the Guidance of

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PARUL UNIVERSITY

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PARUL UNIVERSITY

CERTIFICATE

This is to Certify that Project - 2 -Subject code 203105400 of 7th Semester entitled "Pet Adoption and related services application" of Group No. PIET_CSE_45 has been successfully completed by

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Abstract

Our app is a one-stop-solution for pet owners looking to adopt, connect with other pet owners, seek veterinary advice, and purchase the best products for their furry friends. Our open community allows pet owners to share their experiences, ask for advice, and find support from others who understand the joys and challenges of pet ownership. With the ability to consult with veterinarians and browse a wide selection of pet products, our app makes it easy for pet owners to provide the best care for their pets. Whether you are a first-time pet owner or a seasoned pro, our app has something to offer for everyone looking to build a supportive community for their pets

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Chapter 1

Introduction

1.1 Project problem statement

Despite an increase in pet ownership during the COVID-19 pandemic, India figures show that during the lockdown two-thirds of pet parents had a newfound appreciation for their pets, and 6 in 10 people felt encouraged to adopt one. In spite of this, India's data points to several challenges, such as housing limitations, financial constraints, practical barriers, and low awareness about stray pets, which leads people to buying of breeds instead of adopting from shelters. Based on "State of Pet Homelessness Index" report which includes eight animals' welfare experts from leading organisations around the globe to guide the development of the index, relinquishment levels are higher in India than on a global level with half of current and previous owners stating they have relinquished a pet in the past, compared to 28 per cent on a global level. About 34 per cent said they have abandoned a dog on the streets, and 32 per cent have abandoned a cat. Faced with an overflow of abandoned pets, and an extreme lack of resources to find every pet a forever home, there was a need for innovative solutions to encourage pet adoption..

1.2 Scope

The mobile app will make it easier for people to adopt pets, which will help reduce the number of pets in shelters and increase the number of pets in loving homes. It will also provide pet owners with access to important resources and information to help them properly care for their pets, improving the overall well-being of pets in the community. Many pet owners struggle to find reliable sources of information and support, and this platform aims to fill that gap by providing a space for pet owners to connect with each other, consult with veterinarians, and find the best products for their pets. Additionally, by facilitating pet adoption, this platform can help address the problem of pet homelessness and give animals a loving home. Overall, the goal of this project is to create a

community of pet lovers who are committed to the well-being of their pets and are able to access the resources and support they need to provide the best care possible

1.3 Aim

To create a mobile app that connects pet lovers with other pet owners and shelters, and enable the user to adopt pet instead of buying them and provide them with all the resources required to properly take care of their pets. The main aim is to promote pet adoption and provide all the necessary resources that are needed to make the process more convenient and feasible for the user and enhance the overall experience

1.4 Objective

To create a platform that brings pet owners together and provides them with the resources and support they need to properly care for their pets. Many pet owners struggle to find reliable sources of information and support, and this platform aims to fill that gap by providing a space for pet owners to connect with each other, consult with veterinarians, and find the best products for their pets. Additionally, by facilitating pet adoption,this platform can help address the problem of pet homelessness and give animals a loving home. Overall, the goal of this project is to create a community of pet lovers who are committed to the well-being of their pets and are able to access the resources and support they need to provide the best care possible.

Chapter 2

Literature Survey

2.1 Trends in Mobile Application Development

The actors of the mobile application business are being disturbed by the major software companies. Structural changes have been caused by the newcomers who have imposed and enforced their own rules. Mobile network operators and mobile phone manufacturers are not the only ones who are concerned by the implications of these changes. Additional opportunities and constraints for current mobile application developers are brought about by this changed environment. To take advantage of the current trends, developers need to assess what their options are. In order to explore how structural changes will affect the mobile application development markets, we take a developers perspective. We discuss what aspects developers need to take into account in order to position themselves.

2.2 Android Application Development

Android is a mobile device software stack that includes an operating system, middlew are, and core applications. In collaboration with Open Handset Alliance, Google developed Android, a mobile operating system based on the Linux operating system. Developers can write managed code in a Java-like language that utilizes Java libraries developed by Google, but programs developed in native code are not supported. The Android platform's announcement on November 5, 2007, was announced with the formation of the Open Handset Alliance, a consortium of 34 hardware, software and telecommunications companies dedicated to promoting open standards for mobile devices. Upon release in 2008, most Android platforms will be available under the Apache Free Software and Open-Source License.

2.3 Challenges in Android Application Development:A Case Study

Smartphones are considered to be an important innovation that has changed human life in several aspects. Android It has evolved into the most popular operating system for smartphones. The Android operating system is open source and free Anyone can access it. The Android operating system has many applications (apps) available for your enjoyment and entertainment. and Fierce competition for various exams, students shift from traditional learning methods to M-learning Ready. This white paper covers the design and development of an Android-based app called quiz. Students preparing for competitive exams such as UGC-NET and GATE while on the move. Also featured in this book Various challenges developers face while developing Android apps.

2.4 A Review of Artificial Intelligence Methods for Data Science and Data Analytics: Applications and Research Challenges

The above-mentioned research paper concludes that the Artificial intelligence is a field which requires multidisciplinary expertise where the final goal is to automate all the human activities that presently require human intelligence. Data science is also trending now and analytically deals to solve complex problems. In current world we are advancing more towards artificial intelligence and we are also witnessing the wide spread of it. It is also important to realize the capacity of these technologies in many ways that minimizes the risks safely. Machine learning algorithms are now being used to predict analytics. Deep learning is considered to have more advantages than the conventional machine learning techniques in producing accurate results. The more progress we make in this field, the complexity and the challenges associated with it increases.

2.5 Process discovery on geo location data

The above-mentioned research paper concludes that the Geospatial tracking technology is increasingly becoming one more source of information. Geo location data can be collected in multiple contexts, in a cheap and easy manner. From an operational point of view, this data is used to simply identify the state of a vehicle in terms of positioning and navigation. From a workflow point of view, this data may be used to infer the state of vehicles (or people). An innovative methodology for inferring process information from geo location data is proposed in this paper. Process mining techniques can then be applied to get insight into the processes and operations implicitly described in the geo location data. Complementing the traditional analytical tools for geo location data, the proposed methodology can be used for business intelligence in various

applications such as transportation, logistics and sports. Even though the main scope is the discovery of operations and process behaviours from the geo location data, this methodology may also be used for checking whether the processes are being performed as expected.

2.6 Machine learning for image-based species identification

Accurate species identification is the basis of all aspects of taxonomic research and an integral part of the biological research workflow. Biologists are looking for more efficient ways to meet their identification needs. Intelligent mobile devices, digital cameras, and the mass digitization of natural history collections have created an explosion of freely available biological image data. This rapid growth of biological image data, combined with modern machine learning techniques such as deep learning, offers great potential for automated species identification. Take a look at deep learning neural networks as the technology that has enabled breakthroughs in automatic species identification over the past two years. To stimulate further research in this direction, we provide a brief overview of machine learning systems applicable to species identification problems. We review selected deep learning approaches for image-based species identification and showcase their published applications. This article aims to provide insight into the current state of the art in automated identification and serve as a starting point for researchers willing to employ new machine learning techniques in biological research. Modern machine learning approaches will take time to find their way into the field of species identification, but they argue that in the future there will be a surge in the application of these techniques to problems like artificial intelligence. The system will provide an alternative tool for taxonomic identification in the near future.

2.7 Role of Machine Learning

Investigation and computer modeling of various forms of learning processes is the subject of machine learning. The field of machine learning is now organized around his three main research areas: (1) task-oriented research - development and analysis of learning systems to improve performance on a given set of tasks; Also known as the engineering approach. (2) Cognitive simulation - the study and computer simulation of human learning processes. (3) Theoretical Analysis - A theoretical exploration of the space of possible learning methods and algorithms, independent of application domain. The equally fundamental scientific goals of machine learning include finding different guidance algorithms, the scope and limitations of particular methods, the information that needs to be available to the learner, the problem of handling incomplete training data, and other alternatives. This is a study of learning mechanisms. and general techniques of creation that can be used in many

areas of work

2.8 Cross-platform mobile development approaches

Due to the variety of platforms in the smartphone market, mobile application development is very difficult and very costly. Platform development can be a great solution to this problem. We start by defining cross-platform development, define different types of mobile applications, then define and describe different approaches used in cross-platform development, compare these approaches, and discuss each approach's Allows you to identify strengths and weaknesses. We present the results in a comparison table that considers several criteria. These benchmarks show that choosing a cross-platform development approach depends primarily on two factors: Furthermore, adding that multi-platform tools based on these approaches are somewhat limited (access to various API platforms, performance compared to native applications, etc.), this study goes further than current approaches. It certainly shows that more research is needed to move forward.

2.9 The Importance of User Interface Design

Computer systems (and many other devices, including embedded computers) are It's getting more complicated as technology advances make more things possible. Features that you make available to users. Often there are numerous options and sometimes The kludge way these options are invoked only adds some confusion. User satisfaction, not product satisfaction. This could possibly hinder users Expertise growth. An important factor in the design of complex systems is A good user interface that provides functionality in a way that supports the requirements of Users as learners, not leading to confusion or frustration.

2.10 Smartphone Geo location Technologies

The advent of the Global Positioning System (GPS) has revolutionized the way modern businesses operate and interact with existing or potential customers. Given that all modern digital electronic devices (such as smartphones, laptops, and even digital cameras) have GPS receivers, we can appreciate the potential of using location-based technology. Indoor location systems are part of the location technology that helps people navigate maps indoors. Interfering with location data has been shared for some time, even if people aren't aware of it, but with new laws and regulations passed by commissions like Europe, these apps Forced to turn off geo location tagging by default. However, there seems to be a lack of new research on new geo location technologies, including smartphones such as: B. Visible Light Communication (VLC) for indoor location. Also, important

updates are coming to the new smartphones. For example, the availability of raw Global Navigation Satellite System (GNSS) data. This data has not yet been fully scientifically explored and used to determine how it can positively impact the improvement of location-based systems.

2.11 Adoption and relinquisment interventions at the animal shelter:a review

Each year, nearly 4 million dogs are housed in one of more than 13,000 animal shelters operating across the United States. Review programs run in shelters to increase adoption chances. A shelter dog's morphology plays a large role in choosing an adoptive pup in the kennel, but its behavior also influences its interactions with the adoptive pup outside the kennel. Previous research suggests that dogs have the ability to easily learn new behaviors in shelters, and that programs that improve dog behavior increase adoption rates. While human interactions are well established to improve behavior and physiological outcomes in shelterdwelling dogs, analyzes of the effects of sensory, environmental, and social species-specific enrichment are unequivocal. It does not lead to a conclusion. We also review the literature on transferring owned dogs and the return rate of previously adopted dogs. Although the risks of donation associated with owners and dogs have been debated, there is a significant lack of research on programs that address issues that may prevent dogs from being donated or resonated to shelters in the first place. Factors unrelated to dogs may play a larger role than previously thought. Proposals for further research include cross-site studies, investigations into the effectiveness of shelter enrichment programs, predictive validity of behavioral assessments, understanding adoptive behavior in shelters, and a focus on keeping dogs at home. Includes programs within the community you've put your finger on.

2.12 Increasing adoption rates at animal shelters: a two-phase approach to predict length of stay and optimal shelter allocation

The overall objective of this study is to increase the adoption rates of animals entering shelters by using key factors found in the literature to predict the length of stay. The second phase determines the best shelter location to transport animals using the goal programming approach to make relocation decisions. To accomplish this objective, first, the data is acquired from online sources as well as from numerous shelters across the United States. Once the data is acquired and cleaned, predictive models are developed using logistic regression, artificial neural network, gradient boosting, and random forest. The performance of these models is determined using three performance metrics: precision, recall, and F1 score. The results demonstrate that the gradient boosting algorithm performed the best overall, with the highest precision, recall, and F1 score. Followed closely in second is the random

forest algorithm, then the artificial neural network, and then finally, the logistic regression algorithm is the worst performer. We also observed from the data that the gradient boosting performed better when predicting the high or very high length of stay. Further observing the results, it is found that age for dogs (e.g., puppy, super senior), multicolor, and large and small size are important predictor variables. The findings from this study can be utilized to predict how long an animal will stay in a shelter, as well as minimize their length of stay and chance of euthanizing by determining which shelter location will most likely lead to the adoption of that animal.

2.13 Benefits of having a pet : A review from the science

Egyptian historical records show that the animal has been a companion to humans since ancient times, and that African wild cats have been domesticated for about 4,200 years. Early reports suggesting the usefulness of pets mention a variety of perspectives. Pets functioned as guardians (dogs protect against wolves and bears) and hunting partners (as other human surrogates for interaction). These days, animal companionship is called staying together, eating bread, eating together, and so on. In fact, being close to your pet while making soft, furry meows is an effective solution to your daily stressors. This is true for the millions of pet lovers around the world who have reported significant improvements in their mental, social and physiological health. This article highlighted the results of scientific research on the benefits of owning a pet.

2.14 Application Programming Interface (API) Research: A Review of the Past to Inform the Future

The purpose of this research is to do an API research integration. The research cataloged the academic journal literature on APIs on relevant topics, frameworks, methods, publishing sites, and levels of analysis. The authors draw on a total of 104 articles from scientific journals and conferences published between 2010 and 2018. A systematic literature search was performed on the selected articles. This result suggests that API research is primarily theoretical, focusing primarily on technical aspects such as design and usage. This ignores most of the equally important societal issues such as the business and administrative use of APIs. Identified gaps provide directions for future research.

2.15 Applications of Artificial Intelligence in Machine Learning: Review and Prospect

Machine learning is one of the most exciting things these days. artificial intelligence technology. learning algorithm Many applications that we use every day. one each time Searches use web search engines such as Google and Bing. One of the reasons the Internet works so well is Learning algorithm implemented by Google Microsoft has learned to rank websites. every time Recognize your friends' photos using Facebook, that's right machine learning. Protect your users with an email spam filter Having to sift through a large amount of spam mail, learning algorithm. In this paper, a brief review and future Potential for broad application of machine learning made

2.16 Risk management in software development

Software is today a critical element in every system. The development and implementation of computer programs is influenced by many varied risks which should be managed properly. Risk management in software projects is an activity addressed in several software methodologies, but the different workgroups and organizations apply it in practice in many different ways. A preliminary study has been conducted among Spanish software professionals and developers. It reveals interest in risk management and its techniques, but it also shows that serious deficiencies as well as attitudes which do not help to this activity. These preliminary data represent a basis for a deeper and more detailed analysis of risk management practices which might lead to more effective and practical solutions

2.17 Deep Learning: A Comprehensive Overview on Techniques, Taxonomy, Applications and Research Directions

According to the studies mentioned above, deep learning technology is viewed as a fundamental component of both data science and artificial intelligence. It begins with a history of artificial neural networks before moving on to more modern deep learning methods and innovations in several fields. The main techniques in this field are then examined, along with deep neural network modelling in multiple dimensions. For this, we have also provided a taxonomy that takes into account the various deep learning tasks and their many applications. Further to deep networks for supervised or discriminative learning, we also considered deep networks for unsupervised or generative learning, hybrid learning. In contrast to conventional machine learning and data mining methods, deep learning can create incredibly high-level data representations from vast volumes of raw data. As a

result, it has offered a fantastic answer to a number of real-world issues. In order to be effective, a deep learning technique must have the appropriate data-driven modelling based on the properties of raw data. Before the system can support intelligent decision-making, the sophisticated learning algorithms must first be taught using the information and data linked to the target application that have been gathered. Many applications and research fields, including corporate intelligence, cybersecurity, healthcare, sentiment analysis, visual identification, and many more, are listed in the paper as examples of where deep learning has proven to be beneficial.

2.18 Application Development Using Flutter

According to the data mentioned above, developing cross-platform mobile applications is a top priority in the society and generation we live in today. Developers are forced to choose between creating the same programme multiple times for different OSs (operating systems) or settling for a subpar, similar solution that sacrifices native accuracy and performance for portability. A high-performance and more dependable mobile application can be created using the opensource Flutter SDK for platforms like iOS and Android. Important characteristics of the Flutter include Just-in-time compilation, which runs the computer code that includes compilation at run time rather than before execution, executes the programme. Retailers and small- to mediumsized business owners can benefit from a mobile-driven reward system by using it to draw in more customers, keep those they already have, and encourage them to make larger purchases. Providing loyalty programmes encourages consumers to spend money with your company, leaves a lasting impression on their minds, and puts your brand in the spotlight.

2.19 9 A View of Artificial Neural Network

We came to the conclusion that artificial intelligence is becoming more and more necessary as technology advances due solely to parallel processing as a result of our study of artificial neural networks. The demand for parallel processing is greater now than ever before since it is the only way we can consistently save time and money on tasks involving computers and robots. If we talk about the future work, we can only say that in order to go over the artificial neural network's constraints, we need to create a lot more algorithms and different methods of problem resolution. And if the ideas behind artificial neural networks are merged with those of computational automata and fuzzy logic, we will undoubtedly be able to address some of the shortcomings of this fantastic technology. Why are they so crucial in the artificial intelligence of today? due to the development of intelligent systems, some of which were influenced by biological neural networks. In today's

realm of computer science, ANN offers some highly intriguing alternatives and other applications.

2.20 Comparative Study of Android Native and Flutter App Development

This study compares the steps involved in creating apps utilising the native Android framework versus the cross-platform Flutter framework. There are two sections to this study. The first part uses the Apptim testing tool to run performance tests on two independently designed apps. According to these test cases, native apps perform better in terms of CPU utilisation, memory usage, and app size. A Native app's bundle size is smaller than a Flutter app's. While both programmes consume the same amount of power overall, flutter apps are smoother than android apps in terms of frames per second. The second part of the study consists of an online survey that is carried out by professionals who are currently employed. In the viewpoint of developers, Flutter apps perform better and require less training than Android apps. Yet, because everything in flutter is a widget and creating a navigation drawer, app bar, and bottom navigation requires less work, flutter apps develop faster. On the other side, Android uses XML to design layouts, which are then bound using labour-intensive JAVA code. Our objective for the following research phase is to take usability considerations into account and perform a live app testing survey from end users.

Chapter 3

Project Flow and Methodology

3.1 Project flow

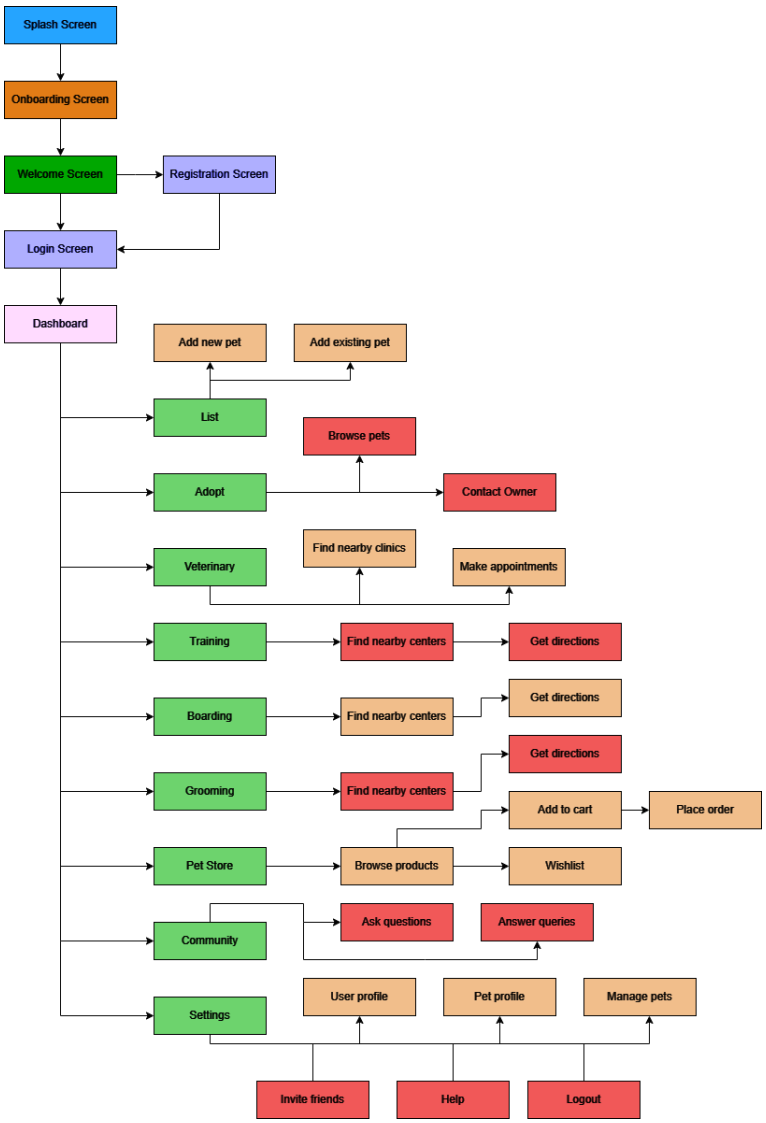


Figure 3.1: Work flow of Application

3.2 Methodology

Our project clearly adheres to the Waterfall Methodology. First, consider why our application employs the Waterfall Model. The Waterfall methodology, also known as the Waterfall model, is a sequential development process that flows like a waterfall through all phases of a project (for example, analysis, design, development, and testing), with each phase completely wrapping up before moving on to the next. The Waterfall methodology is said to adhere to the adage ”measure twice, cut once.” The Waterfall method’s success is determined by the amount and quality of work done on the front end, including the user interface, user stories, and all feature variations and outcomes.

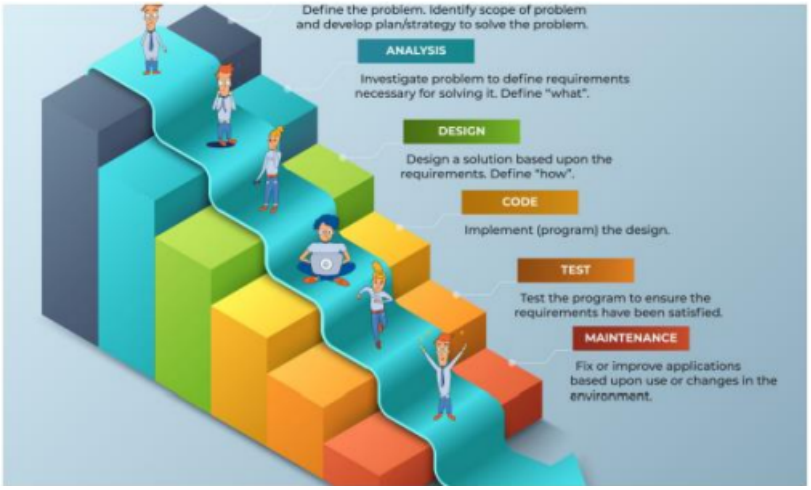


Figure 3.2: Waterfall Methodology

3.3 Benefits of the Waterfall’s organized structure

While the Waterfall methodology may appear to be overly restrictive for some projects, it can be an excellent way to keep a well-defined, predictable project from exceeding time and budget constraints. Clear and detailed organisation can also be beneficial during complex projects involving a large number of people working towards a clearly defined goal.

3.4 Characteristics of Algorithm

The following are the characteristics of an algorithm:

- Input: An algorithm requires some input values. An algorithm can be given either 0 or a value.
- Output: At the end of an algorithm, we will have one or more outputs.
- Unambiguous: An algorithm must be unambiguous, which means that its instructions must be

clear and simple.

- Finiteness: An algorithm must be finite. Finiteness in this context means that the algorithm should have a finite number of instructions, i.e., the instructions should be countable.
- Effectiveness: Because each instruction in an algorithm affects the overall process, it should be effective.
- Language independent: An algorithm must be language independent in order for its instructions to be implemented in any language and produce the same results.

3.5 Factors that we need to consider for designing an algorithm

The following are the factors we need to consider for designing an algorithm:

- Modularity: If we are given a problem and can break it down into small-small modules or small-small steps, which is a basic definition of an algorithm, it means that this feature was perfectly designed for the algorithm.
- Correctness: An algorithm's correctness is defined as when the given inputs produce the desired output, indicating that the algorithm was designed correctly. The algorithm analysis was completed correctly.
- Maintainability: In this context, maintainability means that the algorithm should be designed in a very simple structured manner so that when we redefine the algorithm
- Functionality: It takes into account various logical steps to solve a real-world problem.
- Robustness: It takes into account various logical steps to solve a real-world problem.
- User-friendly: If the algorithm is difficult to understand, the designer will be unable to explain it to the programmer.
- Simplicity: If an algorithm is simple, it is simple to understand.
- Extensible: Your algorithm should be extensible if another algorithm designer or programmer wants to use it.

3.6 Tools and Technology

This section describes about the various tools and technologies used to develop this project.

3.6.1 Flutter

Google announced Flutter in 2017, and it soon rose to the top of the list of the fastest application development platforms. Currently, the third most popular app development framework, Flutter is a high-level native UI platform that is open-source for Android and iOS devices. Flutter is powered by Dart. Flutter's feature-rich user interface makes app development simple for developers. Also, Flutter offers reactive and declarative programming techniques, which makes it comparable to React Native.

3.6.2 Dart

Google released Dart as an open-source, platform-neutral, and object-oriented programming language. With Dart language, you can create apps for both iOS and Android using the same codebase. Dart code can be converted into native code for desktop and mobile platforms. Also, thanks to direct compilation, Dart can connect with any platform thanks to direct compilation without a separate bridge, unlike React Native. Dart is a client-side programming language that offers several tools for creating apps,including design features, dynamic typing, interface, classes, and optional typing. Dart is made to function on both the browser and the server.

3.6.3 Visual Studio Code

Visual Studio Code is a lightweight but powerful source code editor which runs on your desktop and is available for Windows, macOS and Linux. It comes with built-in support for JavaScript, TypeScript and Node.js and has a rich ecosystem of extensions for other languages and runtimes (such as C++, C, Java, Python, PHP, Go, .NET).Visual Studio Code is a streamlined code editor with support for development operations like debugging, task running, and version control. It aims to provide just the tools a developer needs for a quick code-build-debug cycle and leaves more complex workflows to fuller featured IDEs, such as Visual Studio IDE

3.6.4 Firebase

Firebase is a Backend-as-a-Service (BaaS) app development platform that provides hosted backend services such as a real-time database, cloud storage, authentication, crash reporting, machine learning, remote configuration, and hosting for your static files. Firebase supports Flutter.

So, if you add any Firebase plugin to your Flutter app, it will be used by the Apple, Android, and web versions of your app. Running this command ensures that your Flutter app's Firebase configuration is up-to-date and, for Crashlytics and Performance Monitoring on Android, adds the required Gradle plugins to your app.

Chapter 4

Implementation Details

- UI/UX

  We developed the proposed UI/UX design and layout with the help of Figma. Figma is the leading collaborative design tool for building meaningful product design and workflow

- FrontEnd -End Development

  Flutter, Lottie Animations, Rive.

- Back End Development

  Dart, Rest APIs, Google APIs',Postman.

- Database and Storage

  Firebase,Firestore, SQL Lite.

- Integrated development environment(IDE) Android Studio, VS Code.

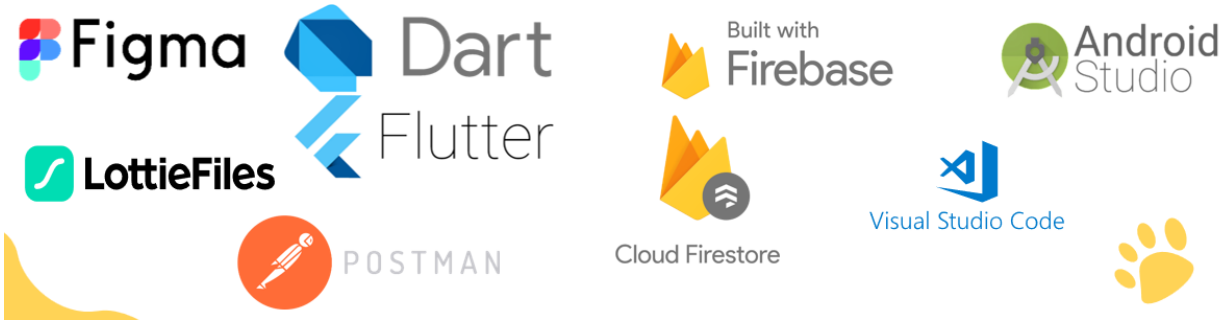


Figure 4.1: Tools

4.1 Sample code

- main.dart

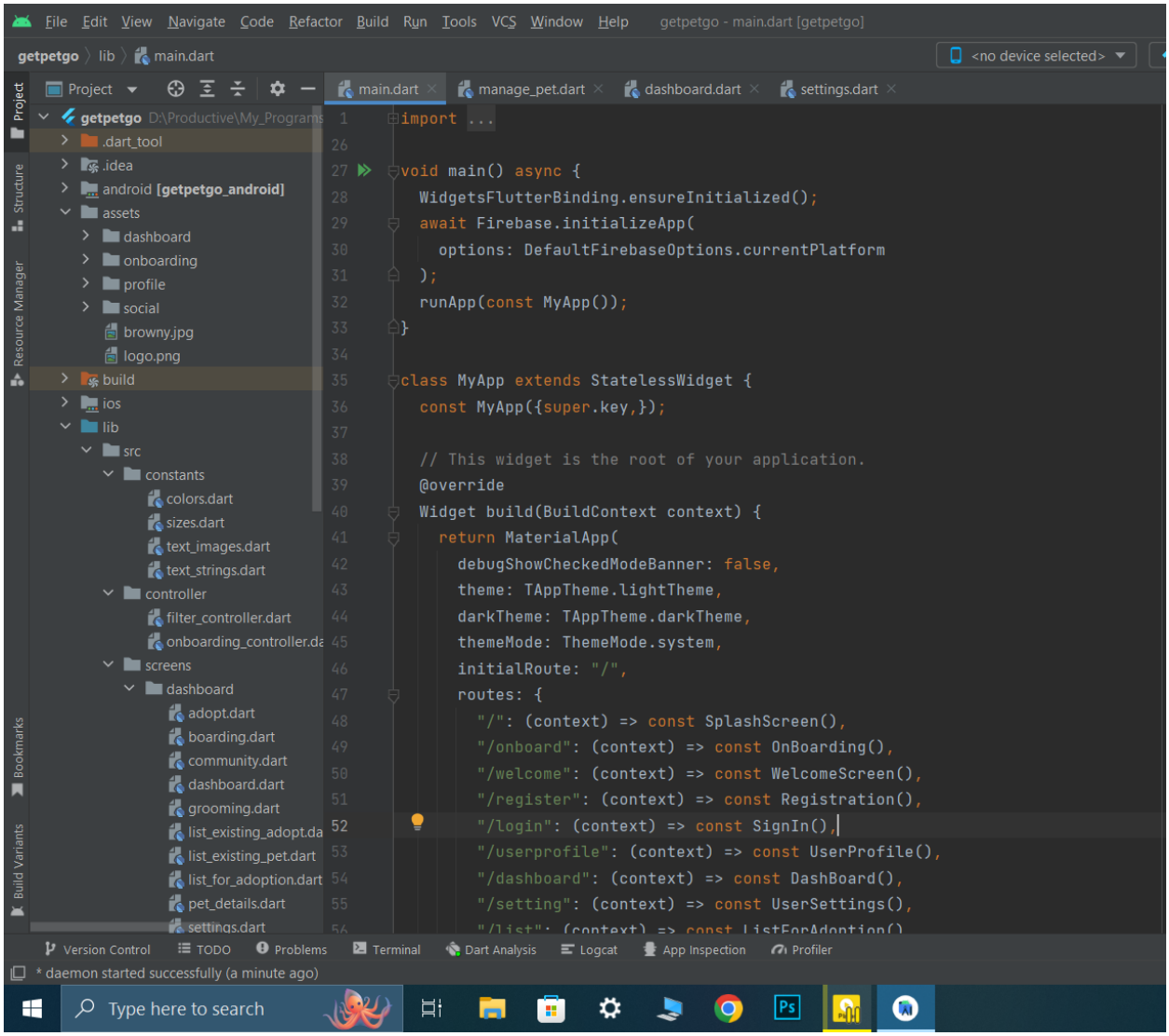


Figure 4.2: Sample code 1

- Filter controller file

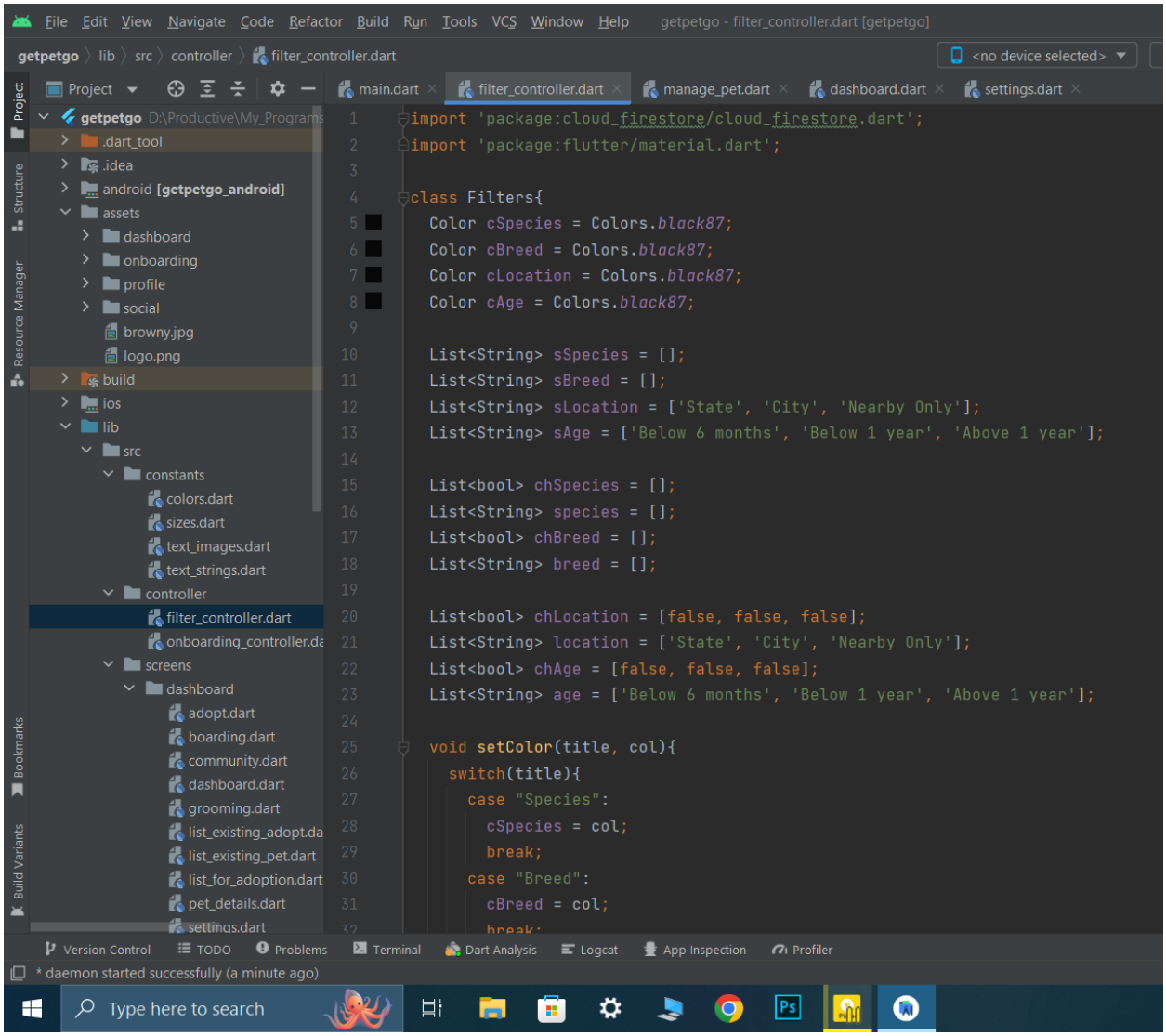


Figure 4.3: Sample code 2

- Login Screen

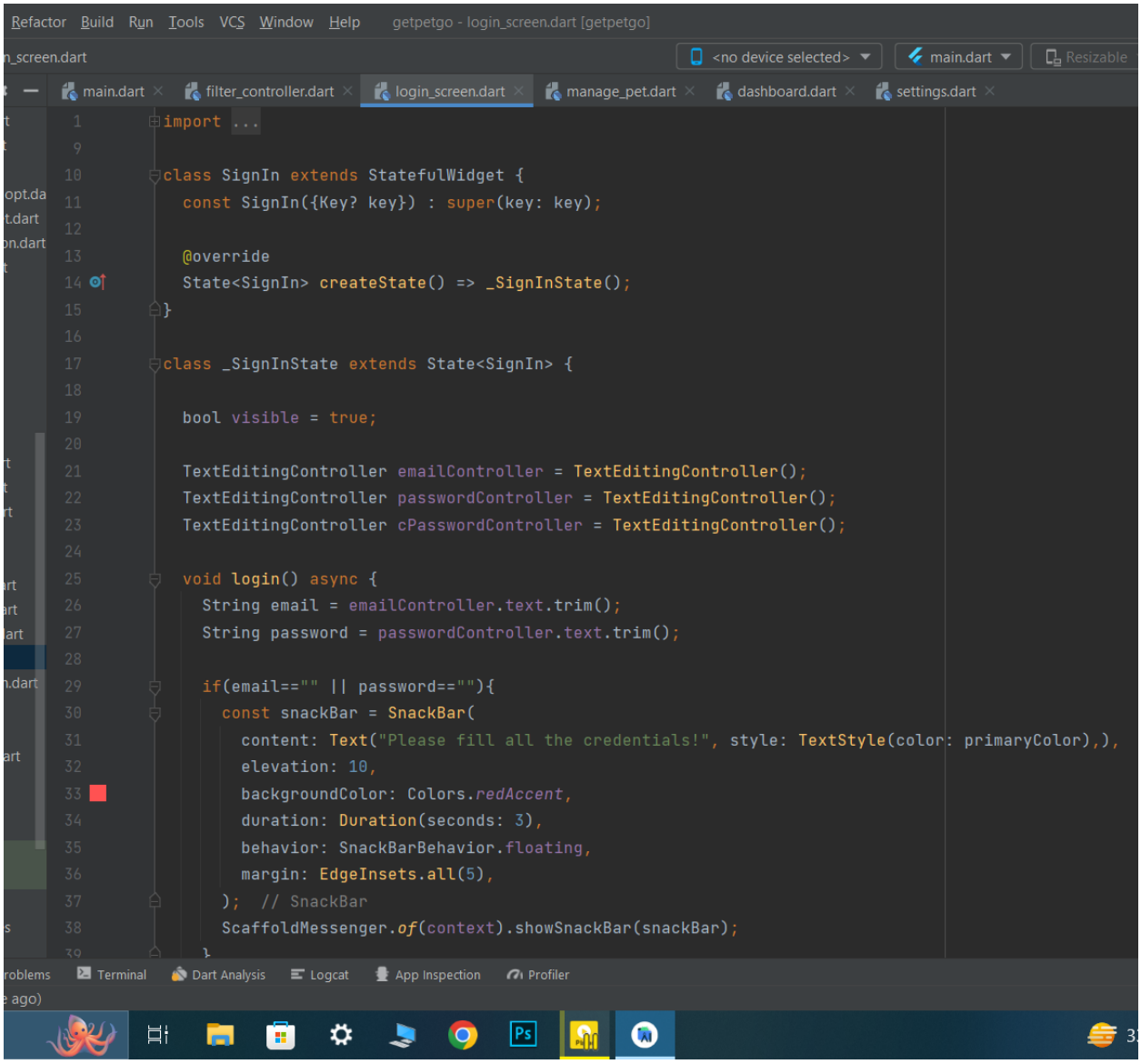


Figure 4.4: Sample code 3

4.2 Testing Cases

4.2.1 Pet Details

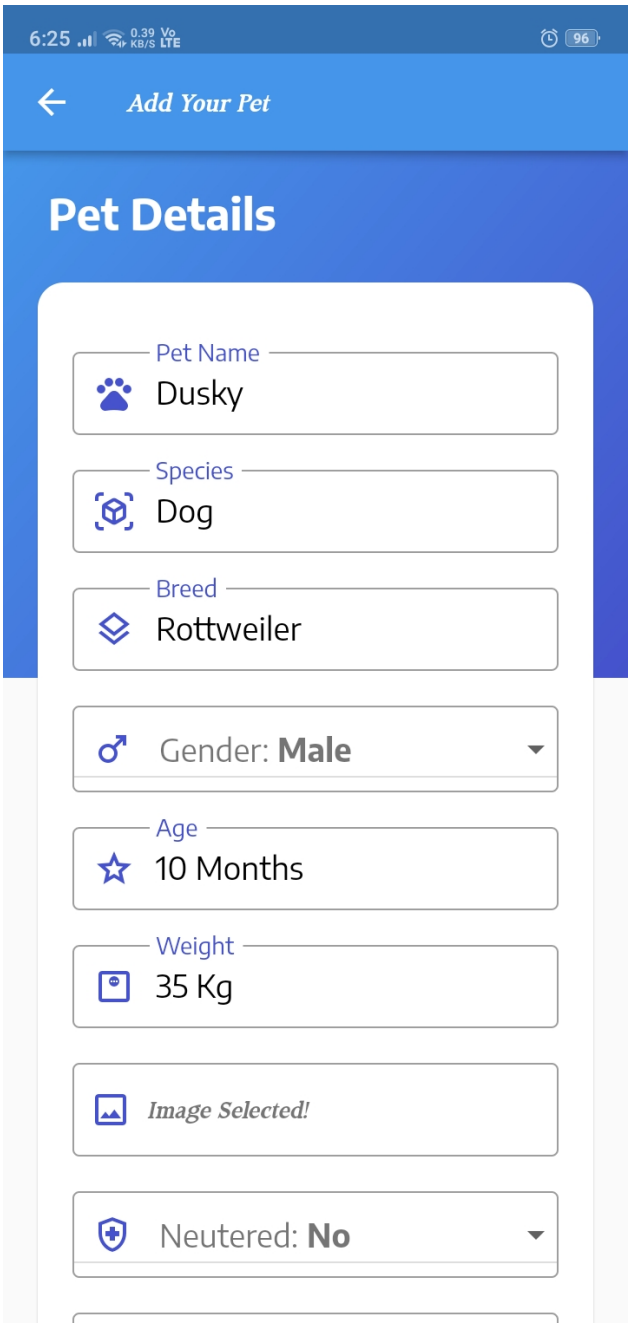


Figure 4.5: Testing Case 1

4.2.2 Adding your pet details

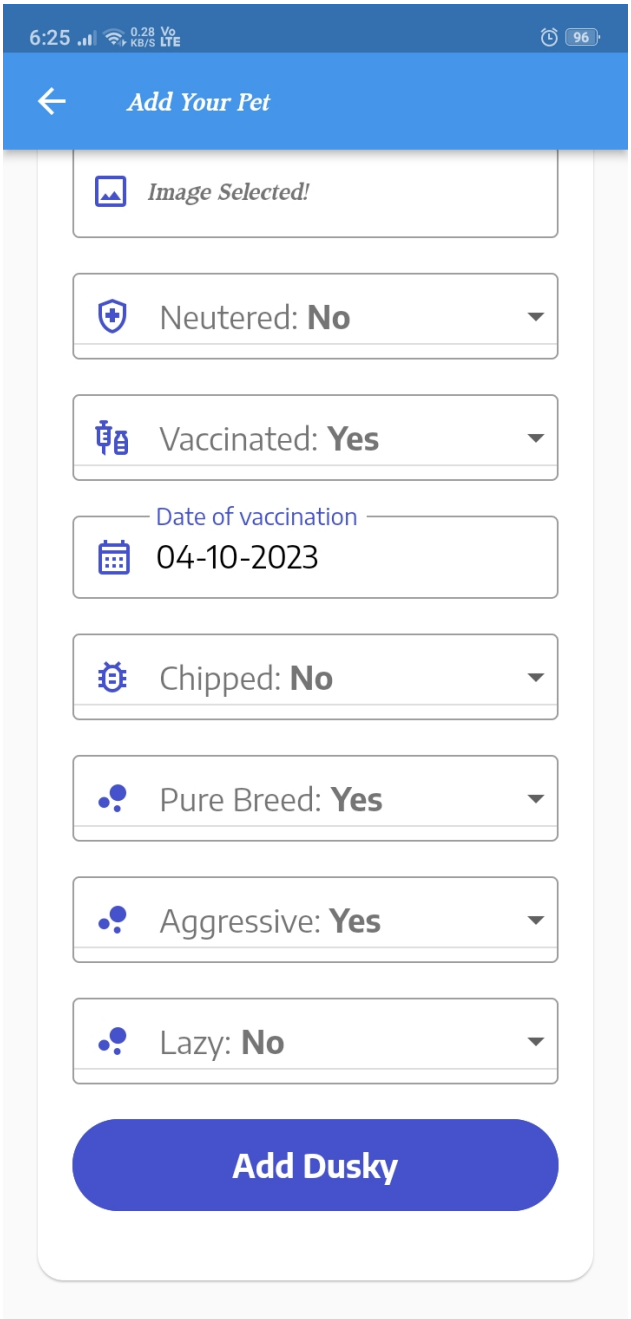


Figure 4.6: Testing Case 2

4.2.3 Select your pet

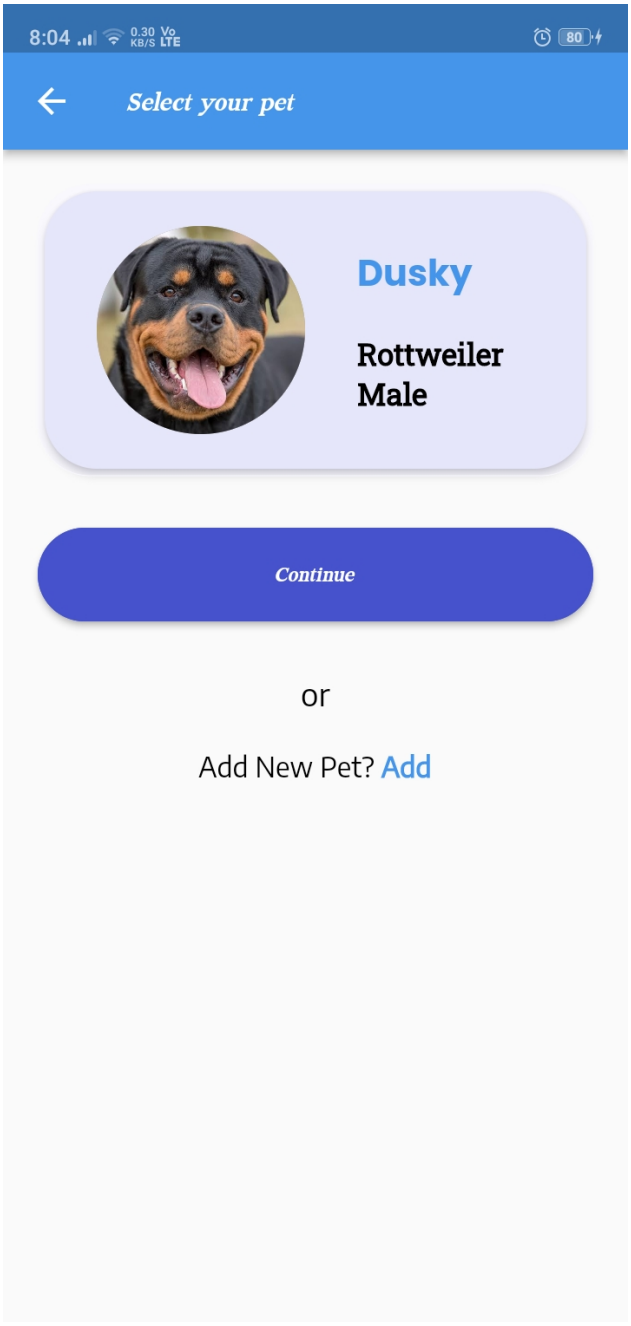


Figure 4.7: add your pet to the list

4.2.4 Pet profile

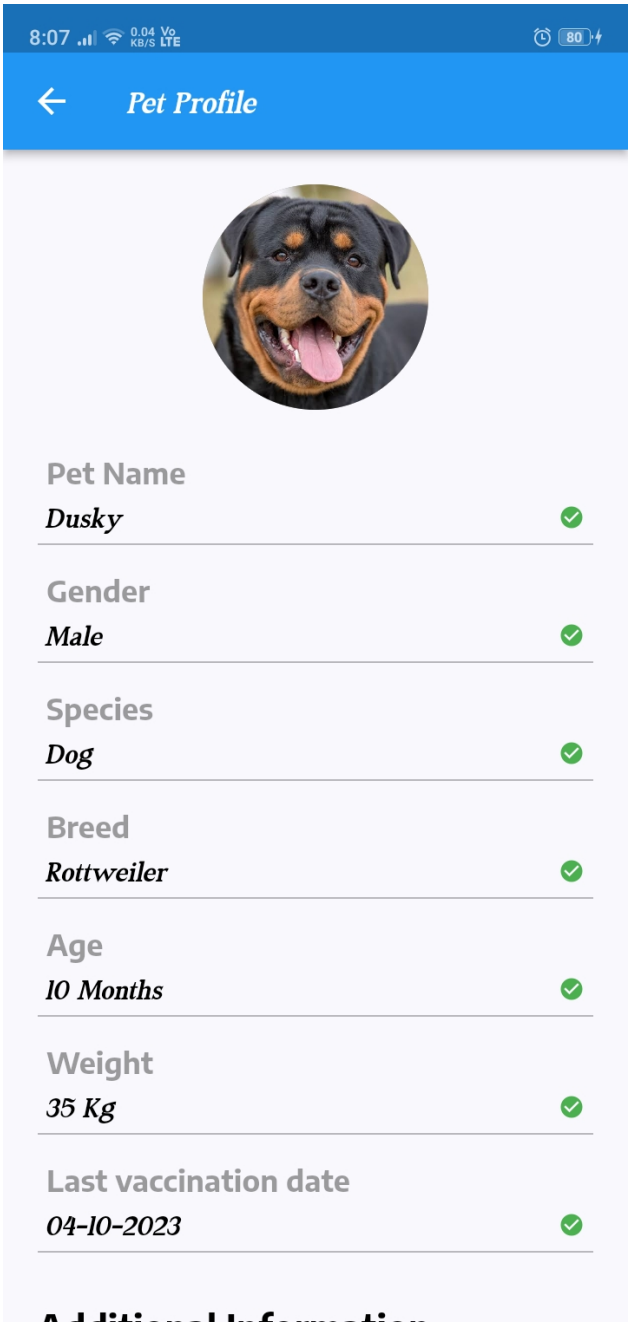


Figure 4.8: your pet full profile

4.2.5 Additional information

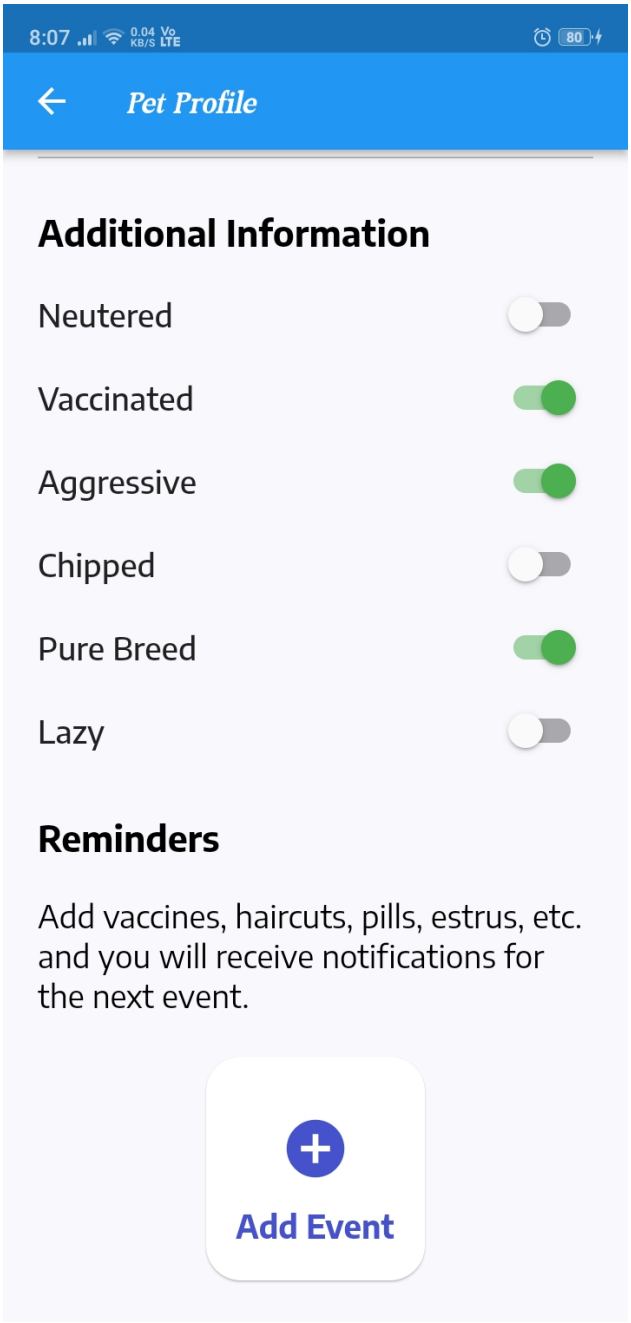


Figure 4.9: Extra Info

Chapter 5

Conclusion

The project has made substantial progress, and the team has demonstrated remarkable dedication, collaboration, and problem-solving abilities in this journey thus far. As we continue to navigate through the remaining of the project, we will apply the lessons learned from the completed segments. We anticipate these efforts will foster greater efficiency, improved problem-solving, and further innovation. The team remains committed to maintaining the high standards of work we have set, ensuring that the completed project meets and possibly exceeds the initial expectations.

This conclusion serves as a checkpoint rather than a closing, a moment to reflect on our achievements and refocus on the tasks ahead. We are optimistic about our future progress and look forward to reporting on our final outcomes in the near future.

Chapter 6

Future Work

6.1 Plans

As of now, we have introduced the basic features needed for pet adoption and listing of pets which was the main goal of our application. The user can list their pet for adoption and their pet would get listed on our application and any other user can look for their new pet among the list of available pets for adoption based on their location. The future enhancement for this project is to introduce a

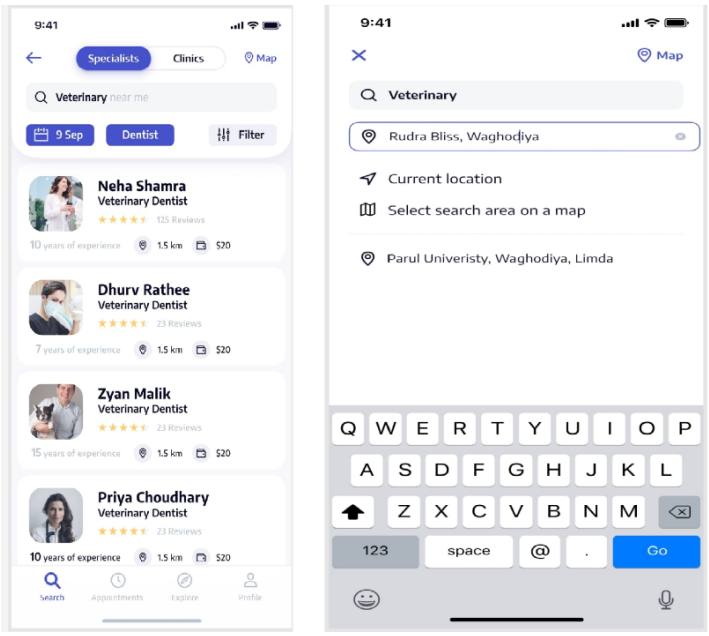


Figure 6.1: Vetinary Mock-up Design

pet store for the user to buy all the products that their pet might need and also implement machine learning for recommending relevant products to user. We also plan to get all the nearby vet clinics with the help of google map and list them in our application so that the user might contact and visit them. We will also be adding some extra features to make the user experience even better and more convenient.

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